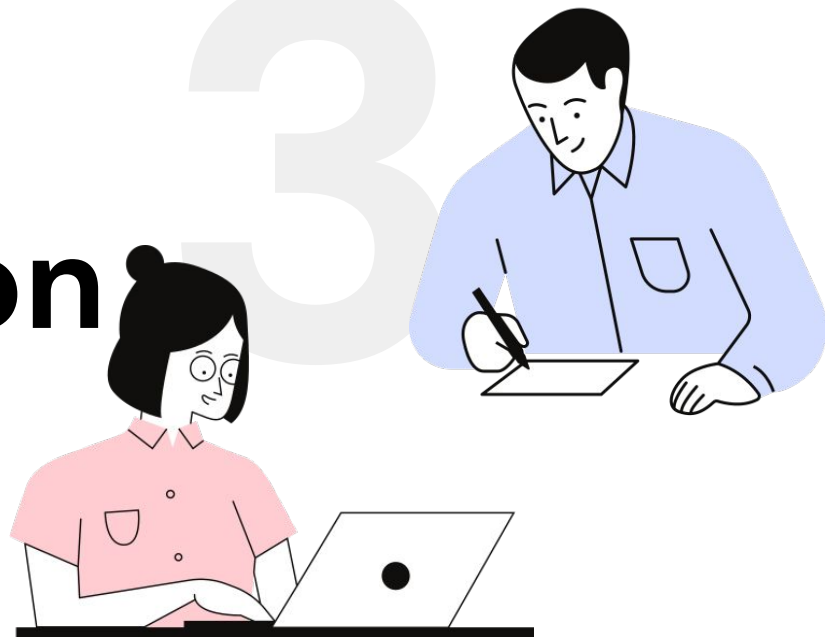




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Objectives

- ❖ Characteristics, uses, media, advantages and disadvantages of storage devices including:
 - Magnetic hard drives: fixed and portable
 - Magnetic tape drives
 - Optical drives including CD, DVD, Blu-ray
 - Fixed and portable solid-state drive (SSD) including:
 - SSD, pen drive, flash drive

Secondary Storage:

- ❖ Non-volatile, larger capacity than primary memory, but slower access.
- ❖ Stores OS, apps, device drivers, and files.

Distinguishing Terms:

- ❖ Media: Physical component where data is stored (e.g., CD).
- ❖ Device: Hardware that reads/writes to media (e.g., CD/DVD reader).

Magnetic Tape Drives

❖ Structure & Function:

- Thin plastic strip coated with magnetic iron oxide layer.
- Data stored as:
 - Magnetized area: Represents a binary 1.
 - Demagnetized area: Represents a binary 0.
- Read/write operations performed by a read/write head.
- Serial access: Data read in the exact sequence it was written.

❖ Data Access:

- Serial access only: Data cannot be randomly accessed; read sequentially.
- Not suited for real-time or online applications: Slow data access speeds.

❖ Primary Use:

- Offline/batch processing: Ideal where quick data retrieval isn't essential.
- High storage capacity: Still used in industrial and academic settings.

Uses of Magnetic Tape Drive

- ❖ **Batch processing** applications: clearing bank checks, utility billing, payroll.
- ❖ **Backup medium**: suitable for storing large volumes of data.
- ❖ **Long-term archiving**: stable and high-capacity, ideal for prolonged storage.

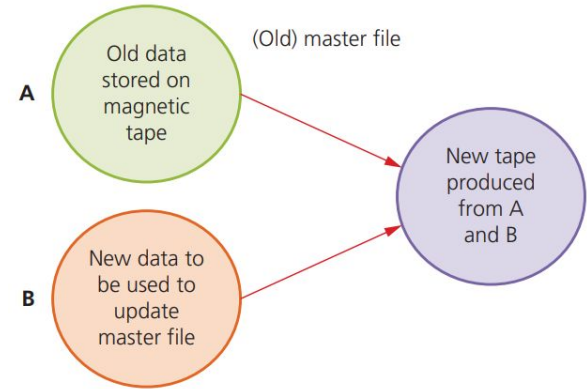


▲ **Figure 3.1** Magnetic tape drive

Advantages & Disadvantages

❖ Advantages

- Cost-effective: Lower cost per byte than HDDs.
- Durability: Stable over time; resistant to deterioration.
- High capacity: Can store vast amounts of data.
- Data transfer rate: Fast transfer speeds, despite slow access time.



▲ **Figure 3.2** Updating a magnetic tape

❖ Disadvantages

- Slow data access time: Requires sequential reading of all prior data to locate specific information.
- Update process is complex: requires a master tape and transaction tape to create a new version, which is slow and prone to errors.
- Susceptible to magnetic fields: data can be corrupted by exposure to strong magnets (e.g., from loudspeakers).

Hard Disk Drives (HDD)

- ❖ Common method of data storage on computers.
- ❖ Stores data in digital format on magnetic surfaces of platters.
- ❖ Often contains multiple platters to increase storage capacity.
- ❖ Direct Access: Allows fast access to specific data without sequentially reading prior data.



▲ **Figure 3.3** Hard disk drive – the hard disk (platter) is the media and the hard disk drive (HDD) is the storage device

- ❖ Key Components:
 - Read/Write Heads: Access data on both surfaces of each platter.
 - Move from center to edge about 50 times per second.
 - Actuators: Move read/write heads; use voice coils similar to speaker electromagnets.
- ❖ Platters:
 - Made from glass, ceramic, or aluminum, coated with iron oxide.
 - Rotate at up to 10,000 revolutions per minute (rpm).
- ❖ Data Storage Layout:
 - Data stored in concentric tracks, each divided into sectors.
 - File Allocation Table (FAT): Manages sector allocation, speeding up data writing.
- ❖ Latency:
 - Delay due to head movement and disk rotation.
 - Notable when accessing frequently or under heavy multitasking, causing delays like "Please wait" or "Not responding."

Fixed Hard Disk Drives

❖ Primary Uses:

- Stores operating system, applications, and working data/files.
- Used in real-time systems (e.g., robotics, plant control) and online systems (e.g., airline booking, stock control).
- Common in file servers for network data storage.

❖ Advantages:

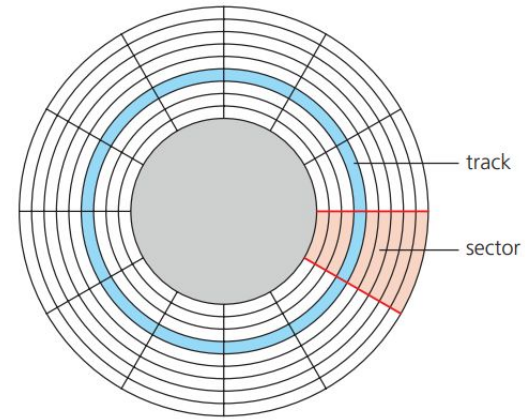
- Fast data transfer rate and access times.
- High memory capacity.

❖ Disadvantages:

- Susceptible to damage if not shut down properly (risk of head crash).
- Many moving parts, affecting reliability.
- Noisy compared to solid-state drives (SSDs).

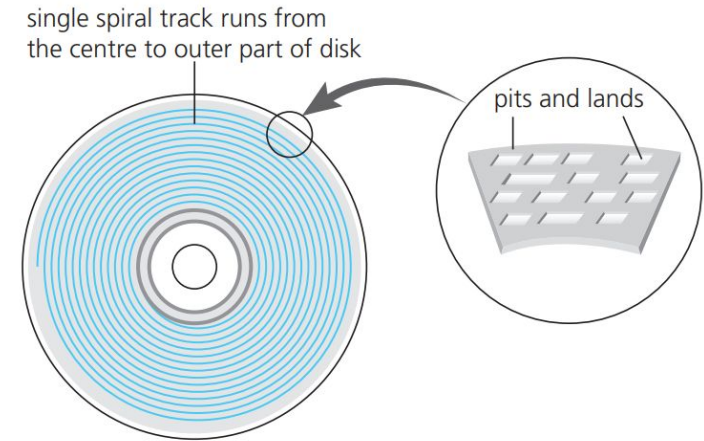
Portable Hard Disk Drives

- ❖ External HDDs connected via USB ports.
- ❖ Serve as backup storage and for file transfer between computers.
- ❖ Uses:
 - Backup systems to prevent data loss.
 - File transfer and software sharing between devices.
- ❖ Advantages:
 - Fast access time and data transfer rate.
 - High capacity, convenient for data mobility.
- ❖ Disadvantages:
 - Easily damaged if dropped or not properly shut down.
 - Slightly slower transfer rates compared to fixed HDDs.



▲ **Figure 3.4** Hard disk platter (showing tracks and sectors)

- ❖ Optical media read/written using laser light.
- ❖ Data stored on a single spiral track running from the center to the edge.
- ❖ Laser reads pits and lands (gaps between pits), converting them into binary data.
- ❖ Three Formats:
 - R: Write-once.
 - RW: Rewritable.
 - ROM: Read-only.
- ❖ Technology:
 - Red laser used for read/write operations.
 - CDs: Laser wavelength = 780 nm.
 - DVDs: Laser wavelength = 650 nm.
 - DVDs have smaller pit size and track width than CDs, offering greater storage.



▲ **Figure 3.5** Optical media

❖ DVD-Specific Features:

- Dual-layering increases storage capacity.
 - Two recording layers joined with a polycarbonate spacer and thin reflector.
 - Laser adjusts focus to read/write each layer.

CD-R and DVD-R:

❖ Uses:

- Home music (CD-R) and movie recordings (DVD-R).
- Data storage for later use or file transfer.

❖ Advantages:

- Inexpensive medium.
- Once finalized, behaves like ROM.

❖ Disadvantages:

- Single-use: Errors render the disk unusable.
- Not universally compatible with all players.

CD-RW and DVD-RW:

❖ Uses:

- Re-recording TV programs.
- CCTV systems.
- Backup device for files/data.

❖ Advantages:

- Rewritable multiple times.
- Supports varied file formats for each use.
- Reduces waste compared to R formats.

❖ Disadvantages:

- More expensive than R formats.
- Prone to accidental overwrites.

CD-ROM and DVD-ROM:

- ❖ Uses:
 - Permanent data storage.
 - CD-ROM: Music files, software, games, encyclopedias.
 - DVD-ROM: Movies, sophisticated games, and large data files.
- ❖ **Advantages:** Cost-effective compared to HDD systems.
- ❖ **Disadvantages:** Slower data transfer and access times than HDDs.

DVD-RAM

- ❖ Operates differently from standard DVD formats.
- ❖ Used as RAM in computers and camcorders.
- ❖ Obsolete and rarely used today.

- ❖ Advanced optical storage media, distinct from DVDs in structure and read/write operations.
- ❖ **Naming:** "Blu-ray" chosen over "Blue-ray" due to copyright limitations.
- ❖ **Technology:** uses a blue laser (405 nm wavelength) versus DVD's red laser (650 nm).
- ❖ **Storage Capacity:** smaller pits and lands enable up to 5 times more data storage than DVDs.
- ❖ Disk Construction:
 - Single-layer Blu-ray: 1.2 mm thick polycarbonate layer.
 - Dual-layer Blu-ray: Two 0.6 mm layers, same as dual-layer DVDs.
- ❖ Data Transfer Rate:
 - DVD: 10 Mbps.
 - Blu-ray: 36 Mbps.
- ❖ **Encryption:** built-in secure encryption system to prevent piracy and copyright infringement.

Features of Blu-ray

Storage Capacity:

- ❖ Single-layer DVD: 4.7 GB (2-hour standard definition movie).
- ❖ Single-layer Blu-ray: 27 GB (2-hour HD movie or 13 hours of standard definition).
- ❖ Dual-layer Blu-ray: 50 GB (4.5-hour HD movie or 20 hours of standard definition).

Interactivity:

- ❖ Recording HD TV programs.
- ❖ Quick navigation to any part of the disc.
- ❖ Playlist creation for movies/TV programs.
- ❖ Editing/reordering of recorded content.
- ❖ Automatic empty space search to avoid overwriting.
- ❖ Access to online content like subtitles.



▲ **Figure 3.7** Blu-ray disc

❖ Applications

- Home video consoles.
- Storing/playing back movies (e.g., 25 GB for a 2-hour HD movie).
- Data storage or hard drive backups on computers.
- Camcorder movie storage (in cartridge form).

❖ Advantages

- Large storage capacity, ideal for HD content.
- Fast data transfer and access speeds.
- Secure encryption system to prevent piracy.

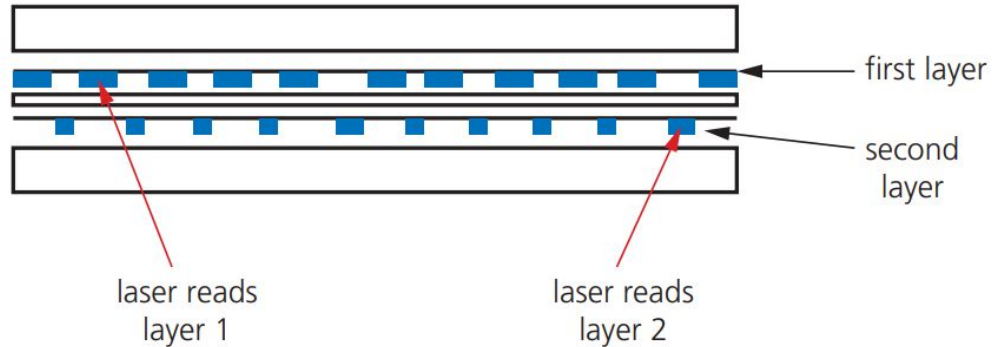
❖ Disadvantages

- Higher cost compared to DVDs.
- Encryption may cause issues when storing video.
- Reduced benefits due to the introduction of HD DVD players.

Comparison Table: CD, DVD, and Blu-ray

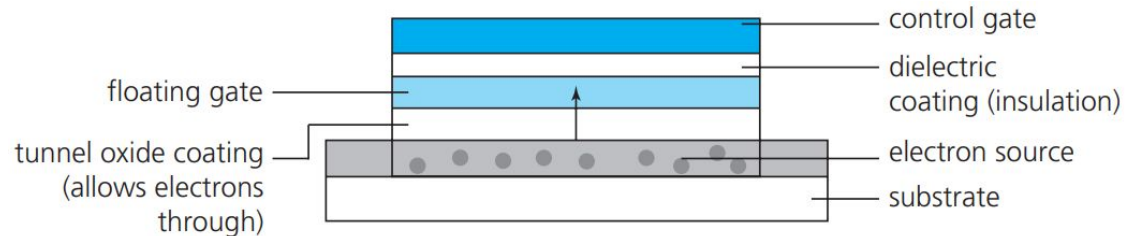
Disk Type	Laser Color	Wavelength	Disk Construction	Track Pitch
CD	Red	780 nm	Single 1.2 mm layer	1.60 μm
DVD (Dual-layer)	Red	650 nm	Two 0.6 mm layers	0.74 μm
Blu-ray (Single-layer)	Blue	405 nm	Single 1.2 mm layer	0.30 μm
Blu-ray (Dual-layer)	Blue	405 nm	Two 0.6 mm layers	0.30 μm

(Note: nm = nanometers, μm = micrometers)



▲ **Figure 3.6** Dual-layering in a DVD

- ❖ Eliminates latency issues found in HDDs by retrieving data at the same rate.
- ❖ Does not rely on magnetic properties and has no moving parts.
- ❖ Data stored in NAND chips as 0s and 1s using tiny transistors:
 - **Floating Gate:** Stores the charge.
 - **Control Gate:** Controls data flow.



▲ **Figure 3.8** Floating gate and control gate (solid-state memory)

Floating Gate & Control Gate Transistors

- ❖ Based on CMOS (Complementary Metal Oxide Semiconductor) NAND.
- ❖ Structure:
 - A grid with floating gates and control gates at intersections.
 - Dielectric coating insulates the floating gate to retain its charge (non-volatile memory).
- ❖ Data Storage:
 - **Charged Floating Gate:** Represents binary 1.
 - **Uncharged Floating Gate:** Represents binary 0.
 - Voltage applied to the control gate traps electrons in the floating gate, defining the bit value.
- ❖ To ensure reliability, devices should be used periodically as stored charge may leak over time (~12 months).

Solid State Drive (SSD)

- ❖ Impact:
 - SSDs have replaced HDDs in many applications, enabling thinner, lighter devices like laptops, smartphones, and tablets.
 - Revolutionized storage due to high speed and reliability.
- ❖ Applications:
 - Storage for files, applications, operating systems.
 - Common in laptops, tablets, smartphones, servers, and cloud storage.

Advantages of SSD

- ❖ Performance:
 - Reliable: No moving parts to fail.
 - Faster access time: 0.1 ms (compared to 10 ms for HDDs).
 - Faster data transfer speed.
- ❖ Energy Efficiency:
 - Lower power consumption.
 - Runs cooler than HDDs.
- ❖ Portability: lightweight and thin, ideal for laptops and portable devices.
- ❖ Instant Readiness: does not require "spinning up" like HDDs.

Disadvantages of SSD

- ❖ Longevity:
 - Limited SSD Endurance:
 - Typically rated for 20 GB of write operations per day over three years.
 - Still not ideal for high-write environments like servers.
- ❖ Cost:
 - Generally more expensive per GB compared to HDDs.
- ❖ Improving Durability:
 - Manufacturers are addressing durability concerns to extend SSD lifespan.

Pen Drives (Memory Sticks)

❖ Overview:

- Small, portable solid-state devices connected via USB.
- Used for file transfer and as a backup medium for music, photos, and other data.
- Draw power directly from the USB connection.

❖ Terminology:

- **Pen Drive:** A flash drive with a USB connector designed for specific tasks.
- **Flash Drive:** Broader term for any solid-state device using NAND technology.



▲ **Figure 3.9** Pen drive/memory stick

Uses of Pen Drives

❖ Uses

- Transporting files between computers.
- Serving as security devices (dongles) to prevent software piracy.

❖ Advantages

- Compact and portable.
- Robust and unaffected by magnetic fields.
- Plug-and-play functionality with most computers.

❖ Disadvantages

- No write protection, making accidental overwrites possible.
- Easy to lose due to small size.
- Risk of data corruption if improperly removed during a read/write operation.

Memory Cards

- ❖ Small solid-state devices used for data storage in portable devices.
- ❖ Common formats include:
 - SD Cards: High-capacity cards for digital cameras, smartphones, tablets.
 - XD Cards: Removable cards for digital cameras.
 - CFast Cards: Used in high-end digital cameras, requiring no battery for data retention.



Uses of Memory Cards

- ❖ Uses:
 - Storing photos in digital cameras.
 - Mobile phone memory cards.
 - Storing music files in MP3 players.
 - Backup storage in handheld computing devices.
- ❖ Advantages:
 - Compact and durable due to lack of moving parts.
 - High data capacity for small size.
 - Easily removable for data transfer to other devices.
- ❖ Disadvantages:
 - Expensive per GB compared to hard drives.
 - Lower capacity and finite read/write cycles.
 - Prone to loss, theft, or damage (especially micro SD cards).
 - Not all computers come with built-in card readers.

- ❖ Decline of CDs and DVDs:
 - Becoming obsolete as many systems no longer include internal drives.
 - USB and flash memory dominate as transfer mediums.
- ❖ Trends in Storage:
 - Cloud storage for global access.
 - Internet-based music streaming replacing physical storage.
 - Increasing popularity of streaming platforms for on-demand content.
- ❖ Smart TVs:
 - Serve as central devices for internet-based movie and TV show streaming.
 - Reduce the need for physical DVD or Blu-ray players.